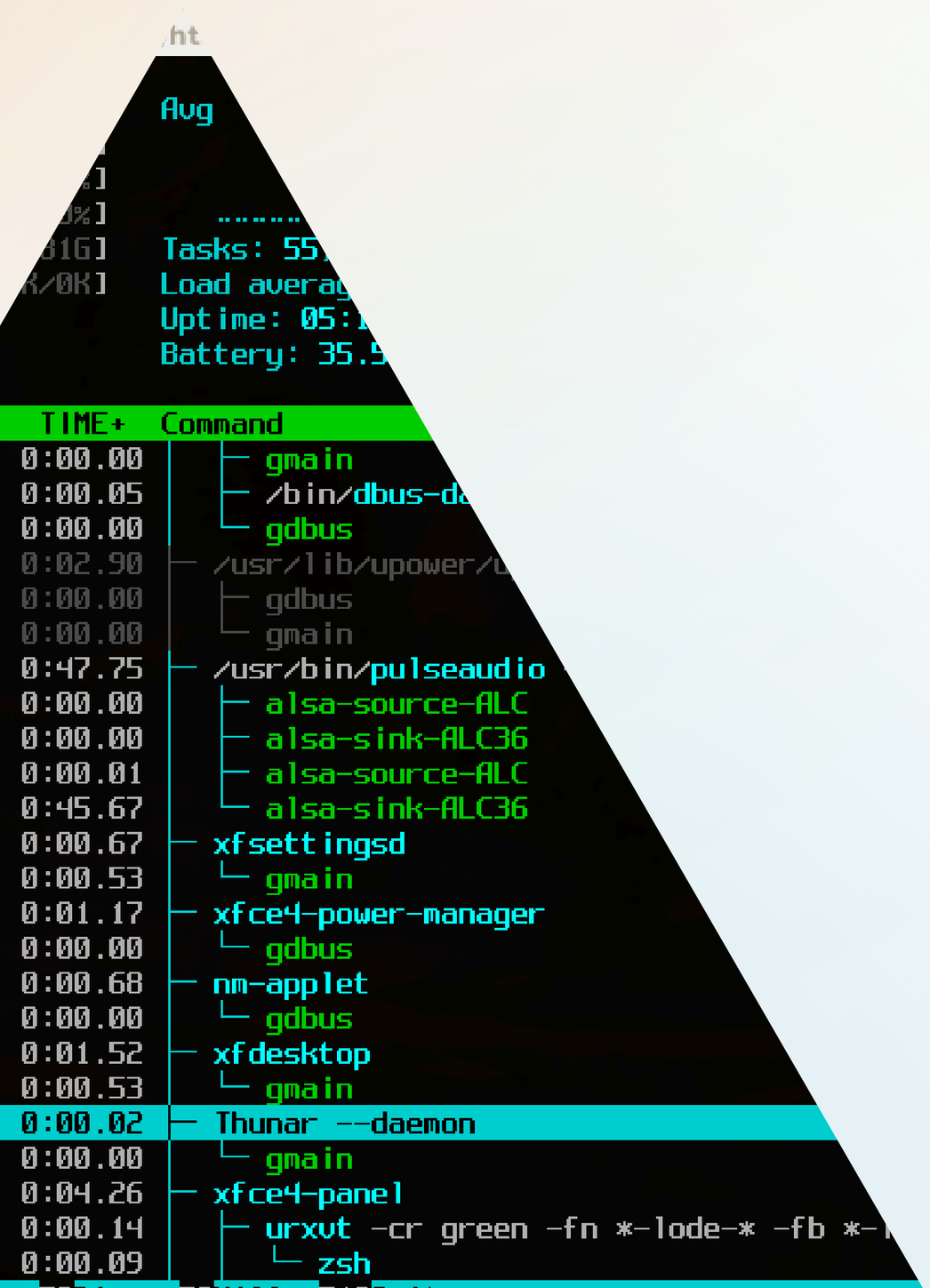

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3

Commanding the Machine

Team Newton

System & AI Essentials

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Join the lecture online on your dashboard.

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Activity

Open Finder

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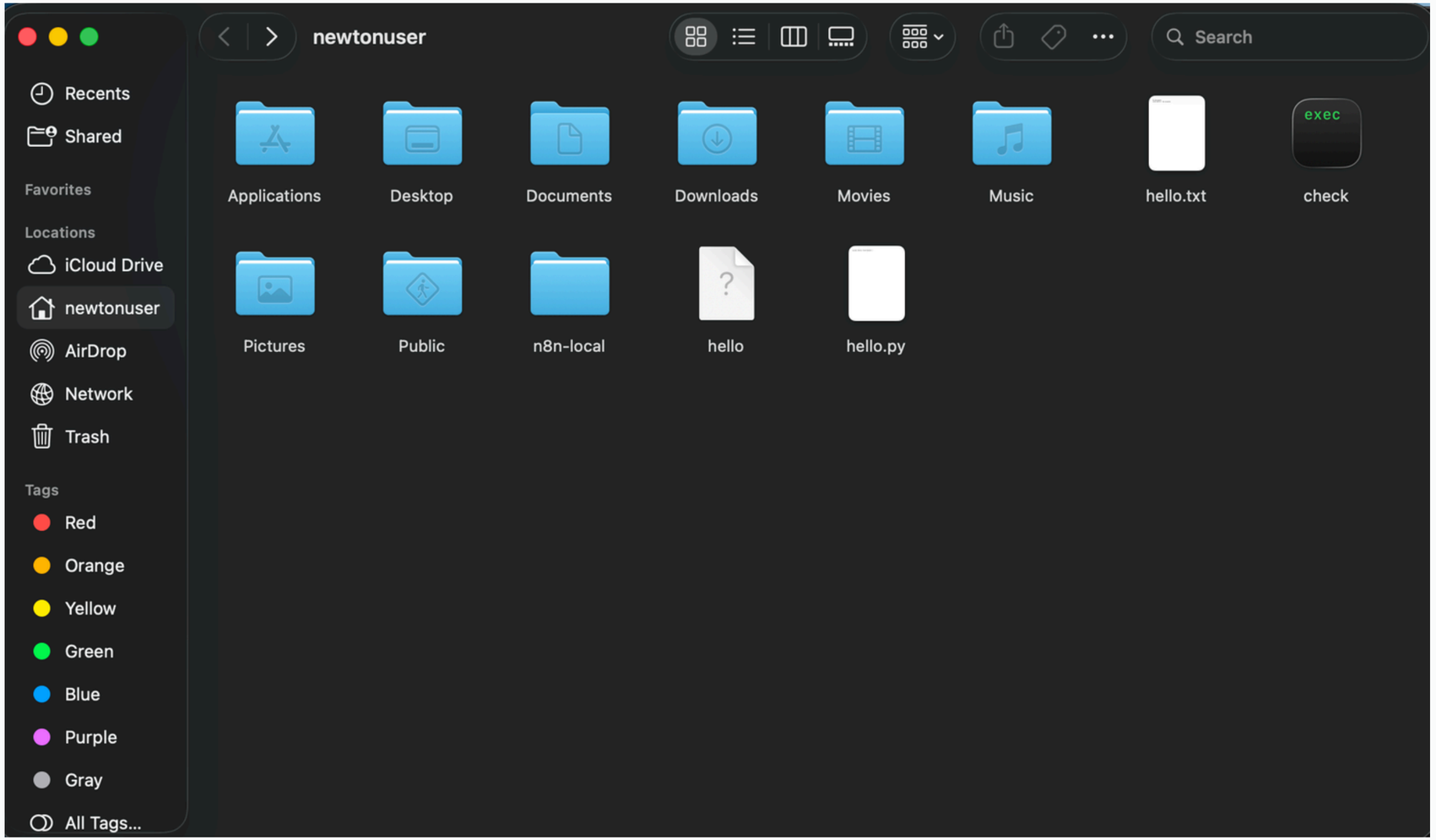
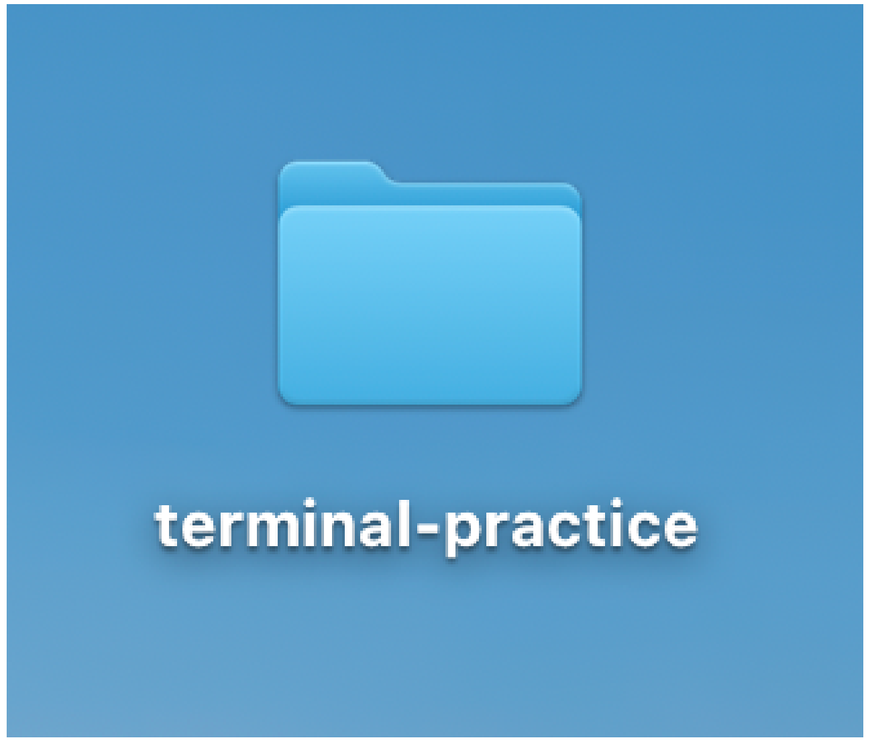


Create a folder on your desktop

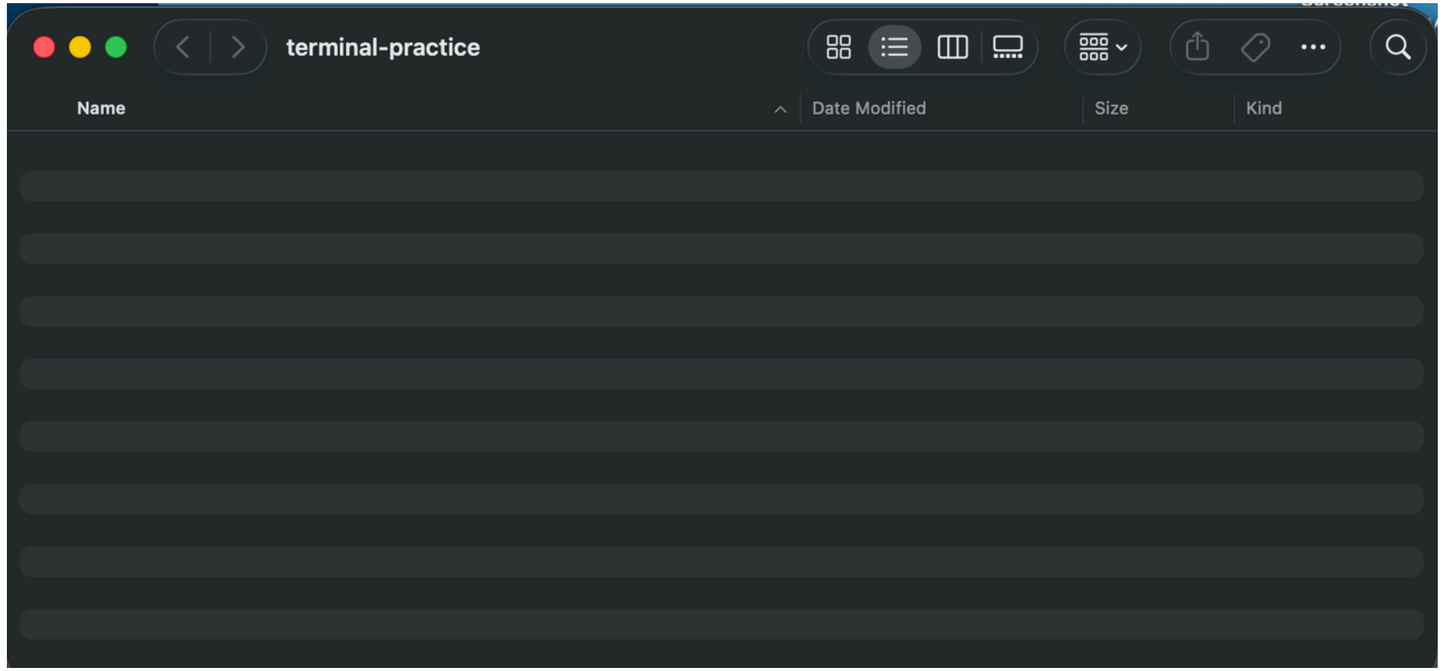
folder name : **terminal-practice**

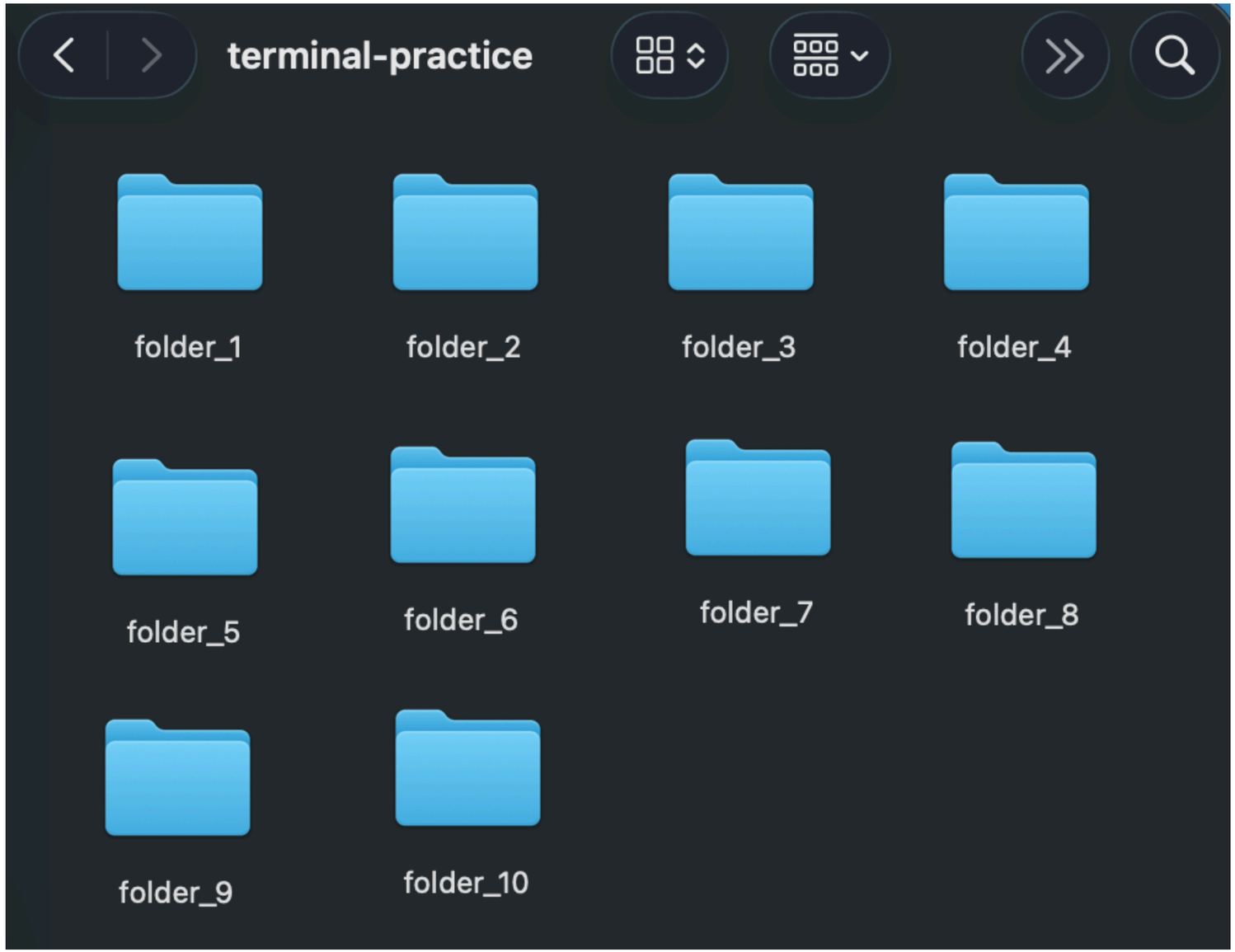


Folder challenge

Inside **terminal-practice** create 10 folders :

- folder_1
- folder_2
- folder_3
- folder_4
- folder_5
- folder_6
- folder_7
- folder_8
- folder_9
- folder_10




terminal-practice
folder_1
folder_2
folder_3
folder_4
folder_5
folder_6
folder_7
folder_8
folder_9
folder_10

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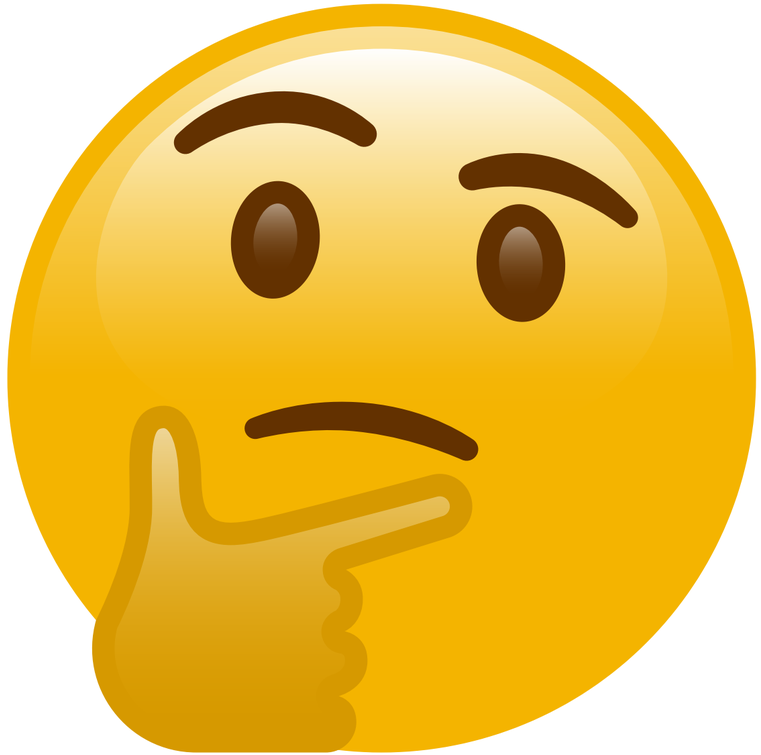
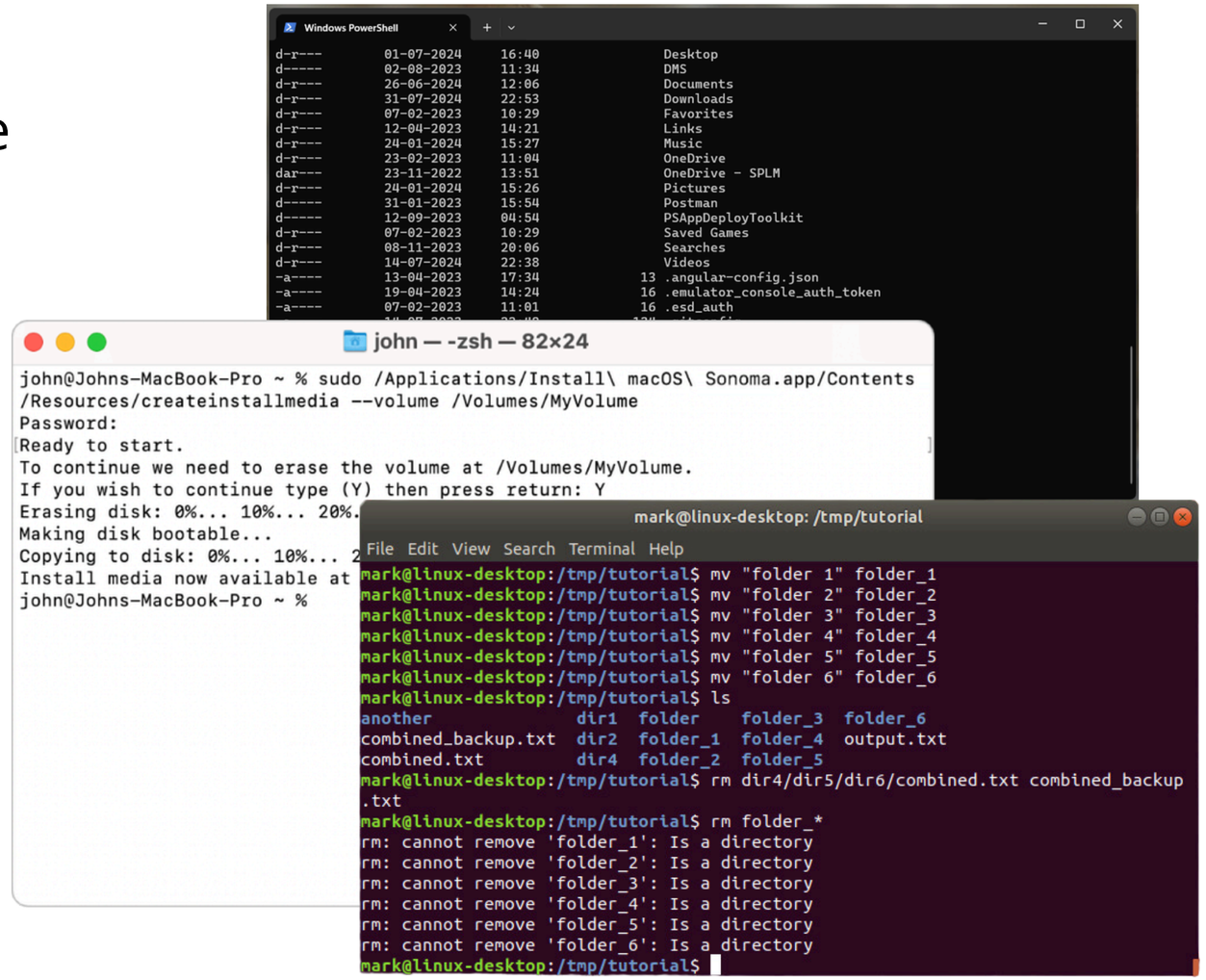
How long do you think it would take you to create 1000 folders this way?

Is there any shortcut?



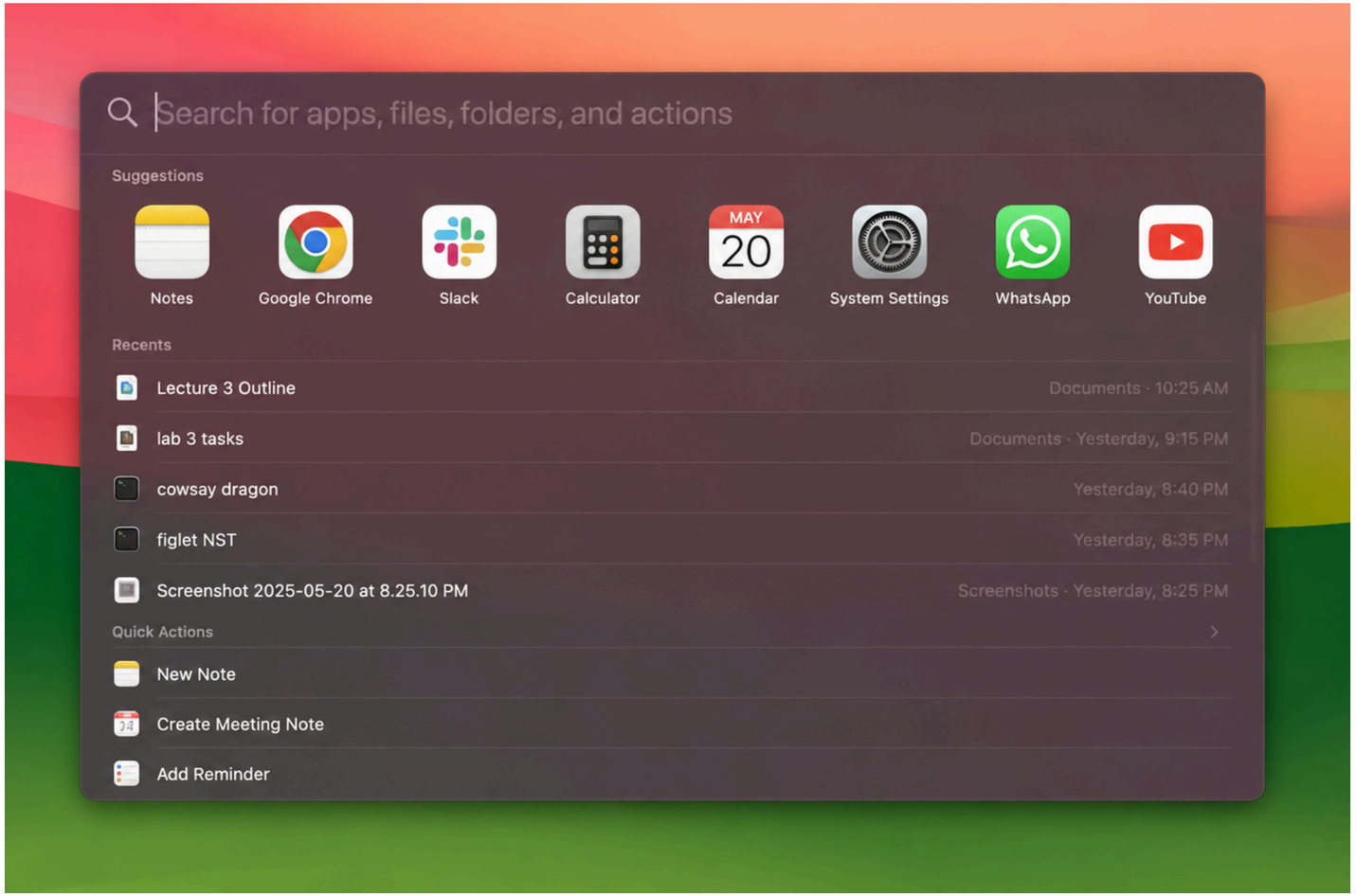
The Command Line

Command Line Interface (CLI) is a type of **user interface** that allows users to interact with a computer system by **typing commands** into a text-based interface.



Open Terminal

1. Press ⌘ Command + Space
2. Type Terminal
3. Press Enter



Terminal

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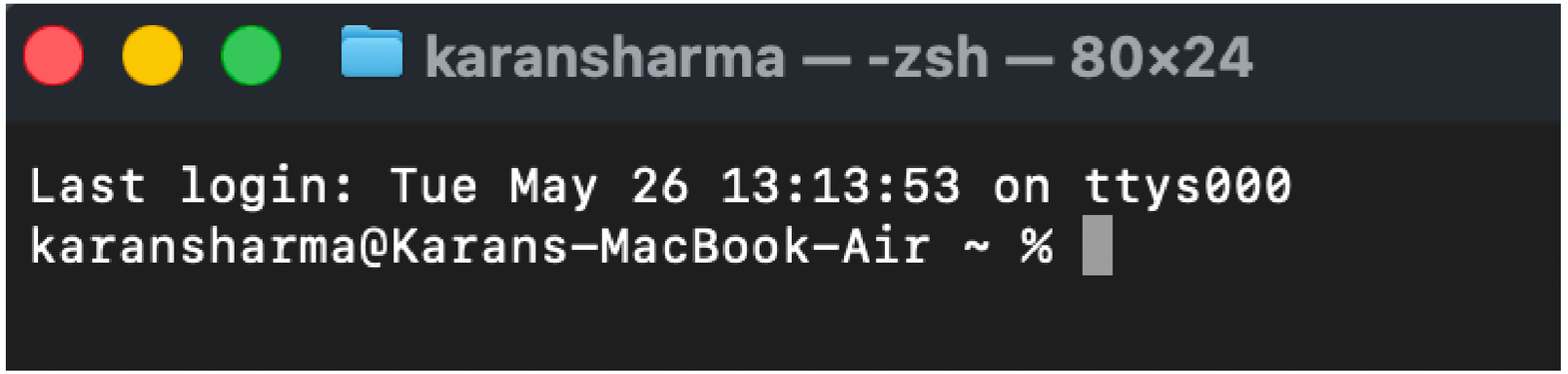

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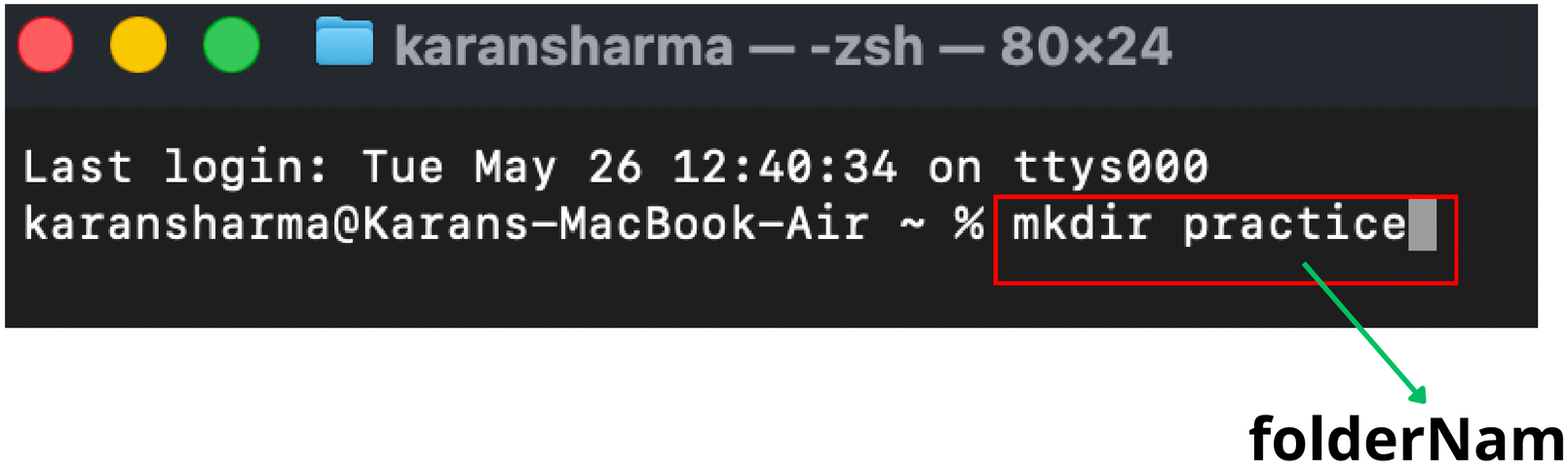
Today’s First Terminal Command

mkdir — Make Directory

Creates a new **directory (folder)** directly from the terminal.



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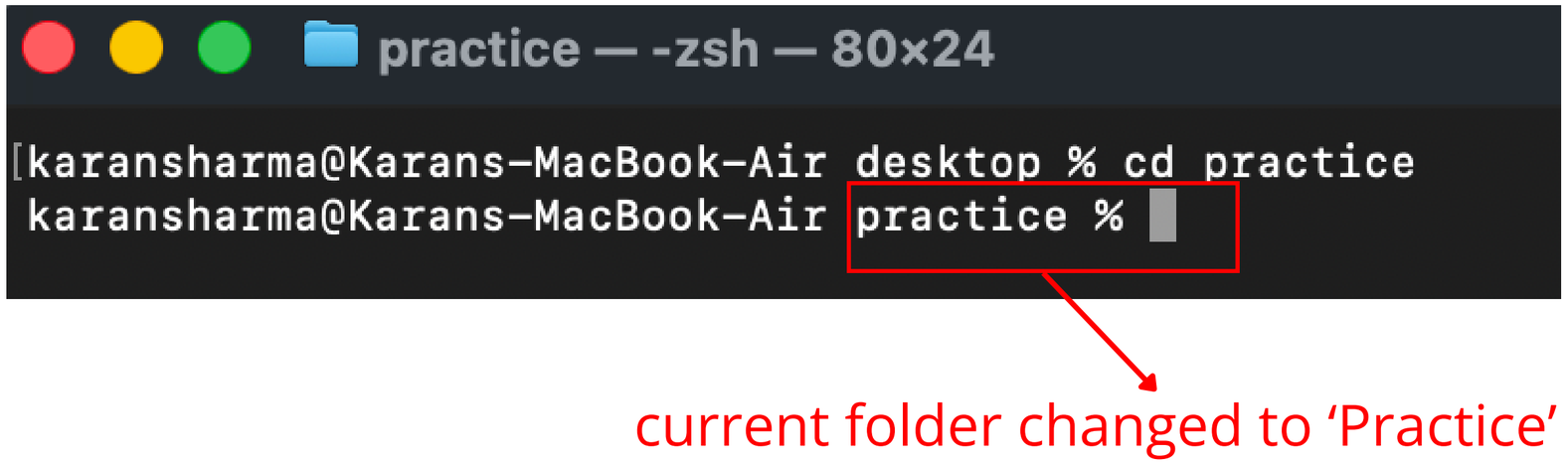
How to move from one folder to another?

cd - change directory

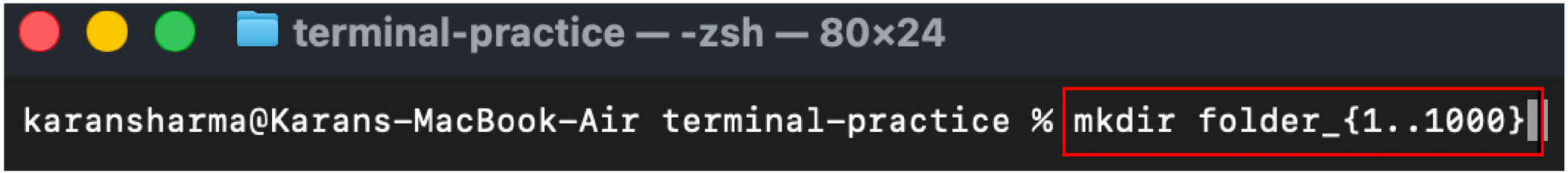
cd is like telling your computer:

🗣️ *"Take me into that folder."*

It lets you **move from your current folder to another one** — like walking into a different room in your house.



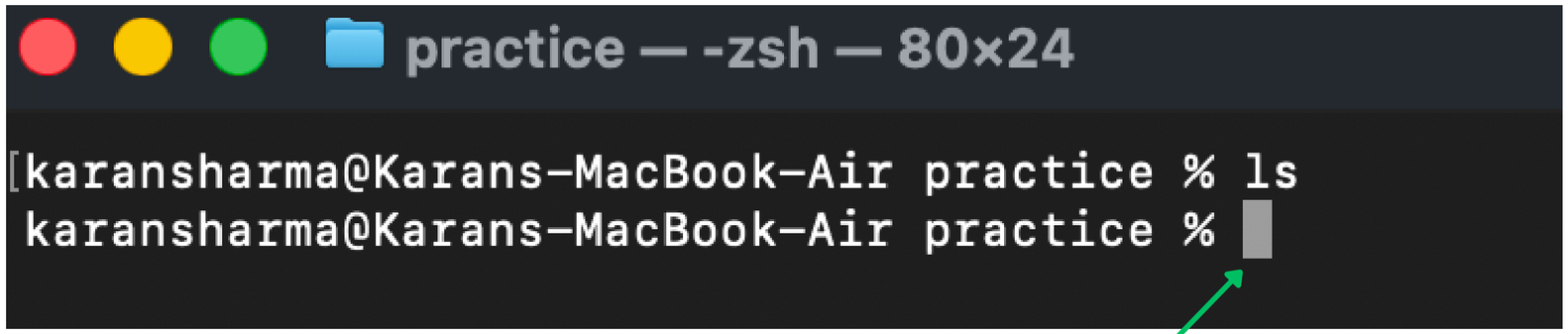
One Command - One thousand folders



We create a folder. How do we see what is inside it?

ls : list files and folders

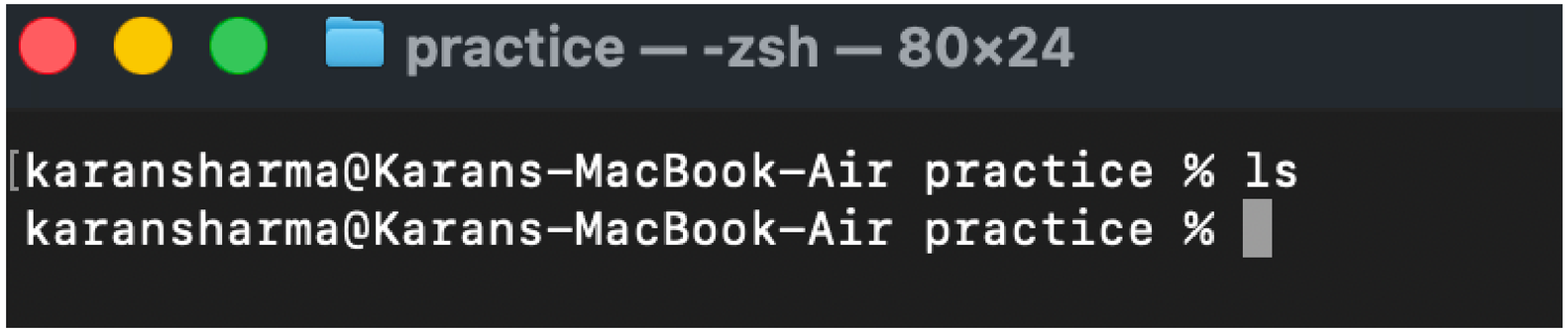
Shows the files and folders present in the current location (directory).



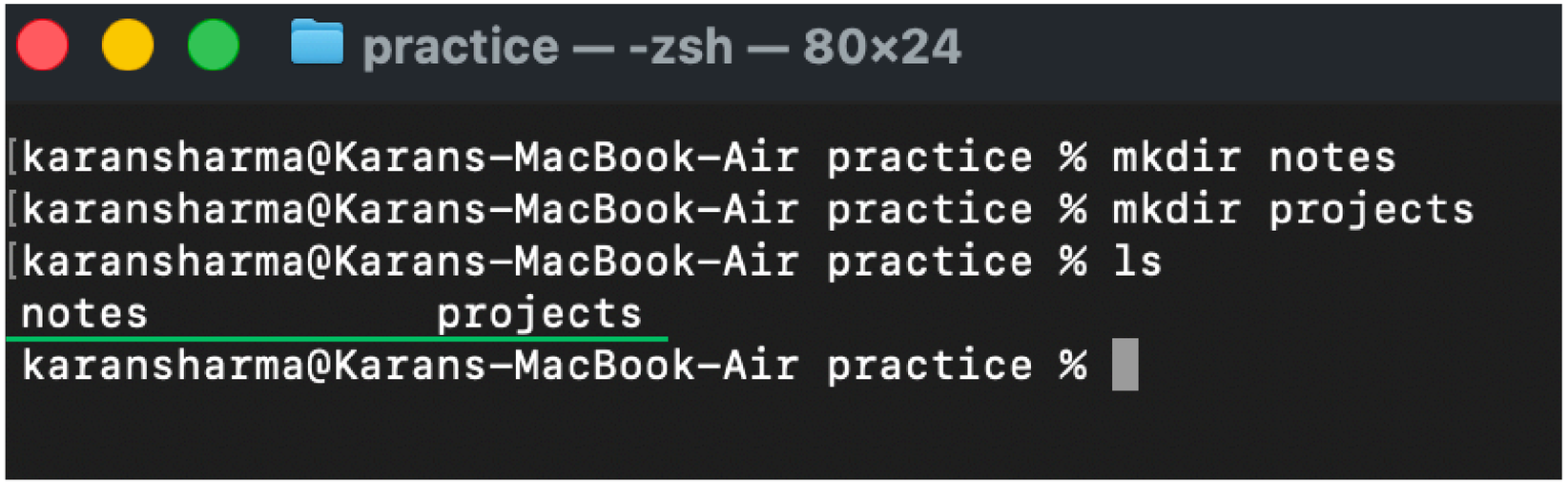
Nothing comes up. Why?

The Folder is Empty

The practice folder is currently empty.
ls only shows files and folders that exist inside the **current directory.**



Let’s create some folders inside practice and try ls again.


practice — -zsh — 80×24
karansharma@Karans-MacBook-Air practice % mkdir notes
karansharma@Karans-MacBook-Air practice % mkdir projects
karansharma@Karans-MacBook-Air practice % ls
notes projects
karansharma@Karans-MacBook-Air practice %

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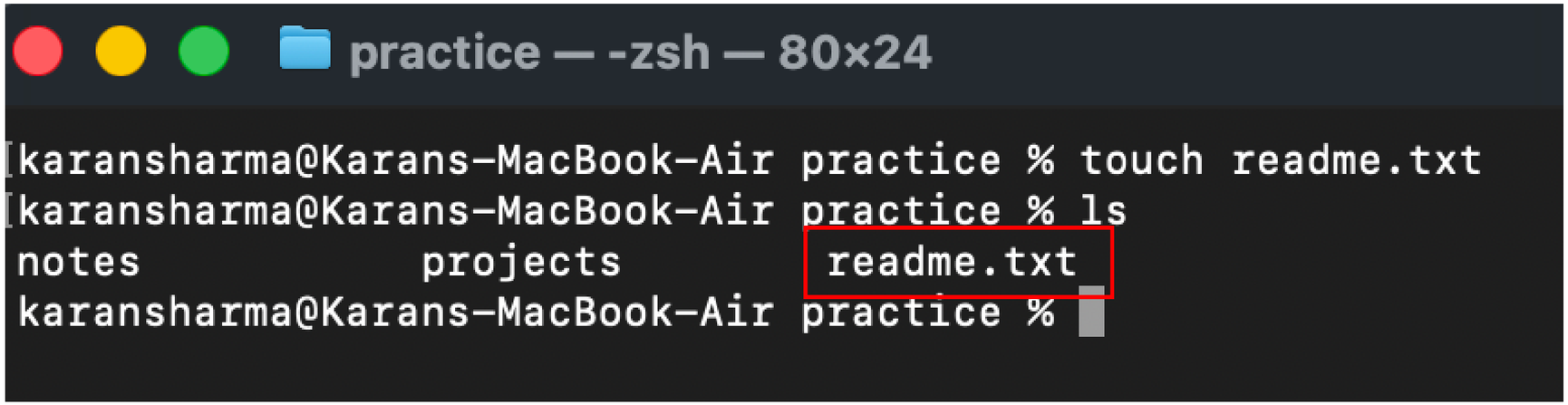
We have only folders. How to create a file?

touch : create a file

Creates a **new empty file** directly from the terminal.



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We have a file. But it is empty. How do we put something in it?

echo : Print lines

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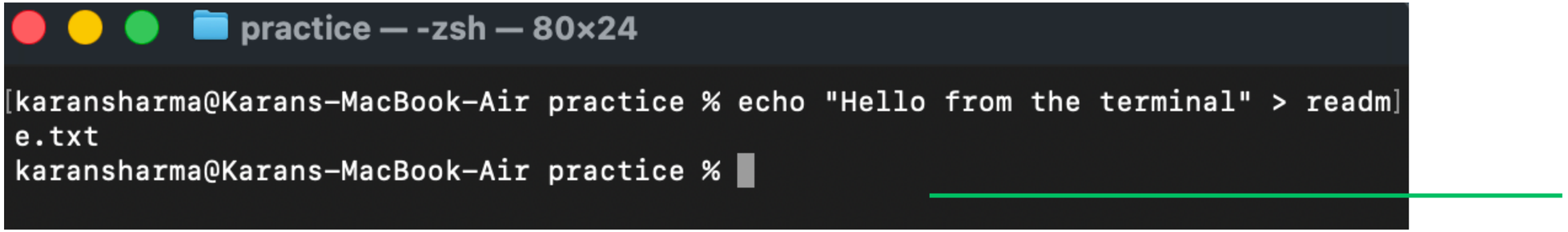
Used to add text into a file directly from the terminal.



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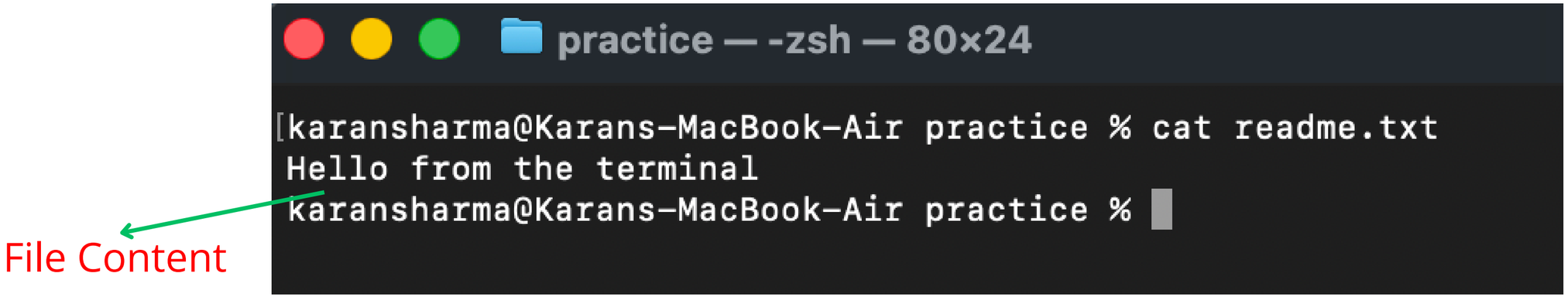
How do we read the content of the file?

cat : read a file

Displays the content of a file directly in the terminal.



Where Are We?

We created folders, moved into folders, created file, but now imagine this:

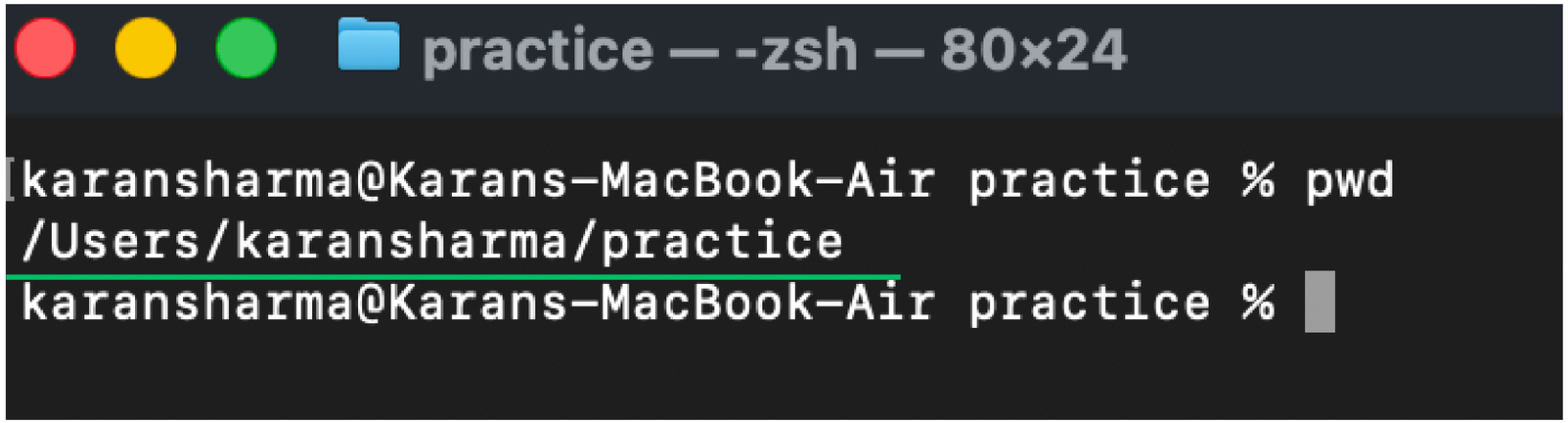
If I close my eyes right now and ask -

Where exactly are we inside the computer?
Could you answer that?

pwd : print working directory

Shows the current location (directory/folder path) you are working in inside the terminal.

Absolute path



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Advanced File and Directory Management

cp, mv & rm

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The cp command **copies files or directories** from **one location to another**, preserving the original files.

The mv command **moves files or directories to a new location** or renames them, effectively changing their path or name.

The rm command **removes files or directories**, permanently deleting them from the filesystem.

cp, mv & rm: Usage

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```
touch kitkat.txt

cp kitkat.txt perk.txt

mv perk.txt munch.txt

rm kitkat.txt
rm munch.txt

```

cp: Example: Copy Single File

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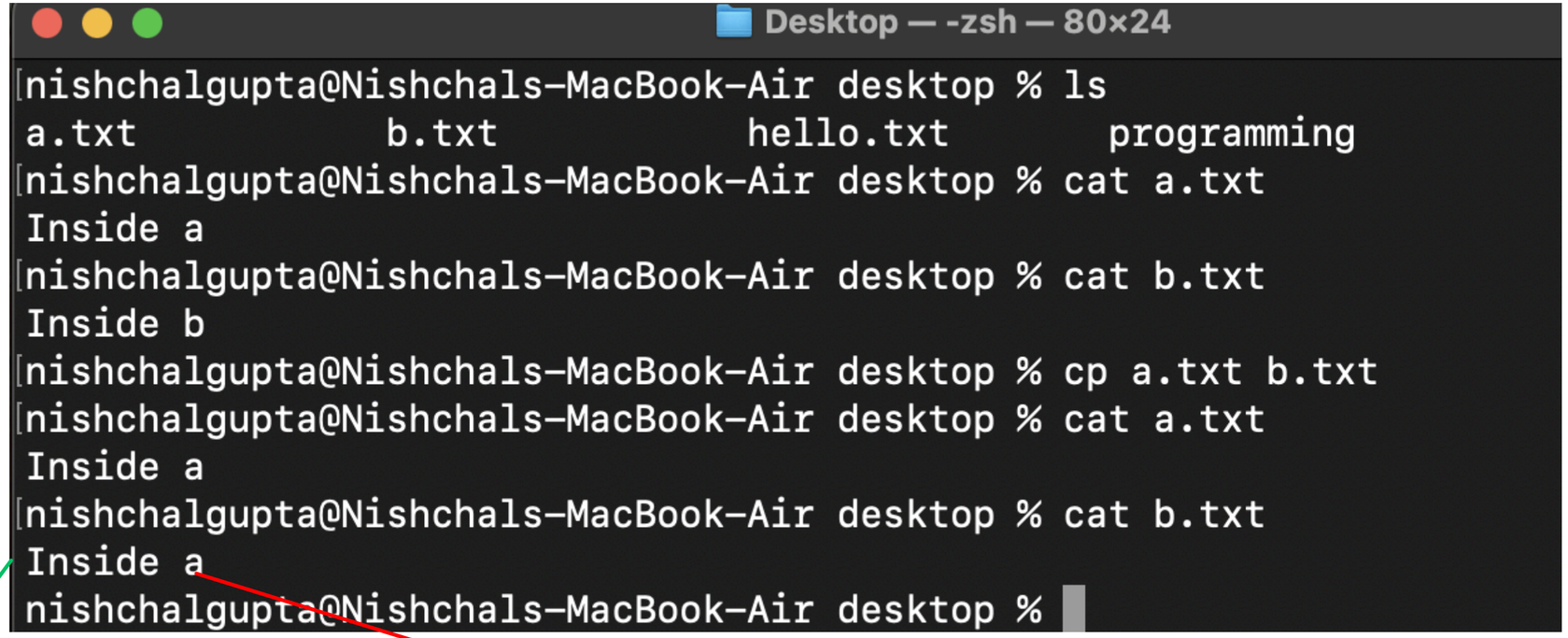

b.txt is overwritten with whatever is in **a.txt**

file **b** now contains the content of **a**

mv: Example: Move Files and Folders

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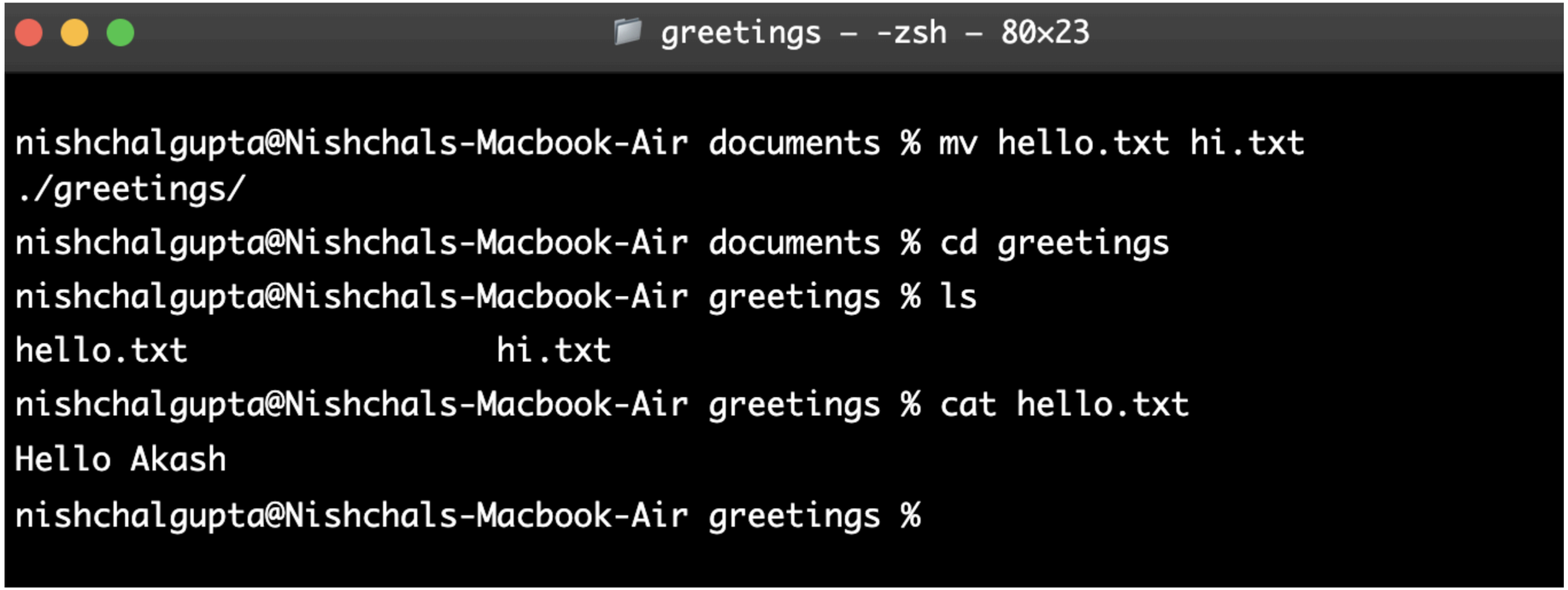
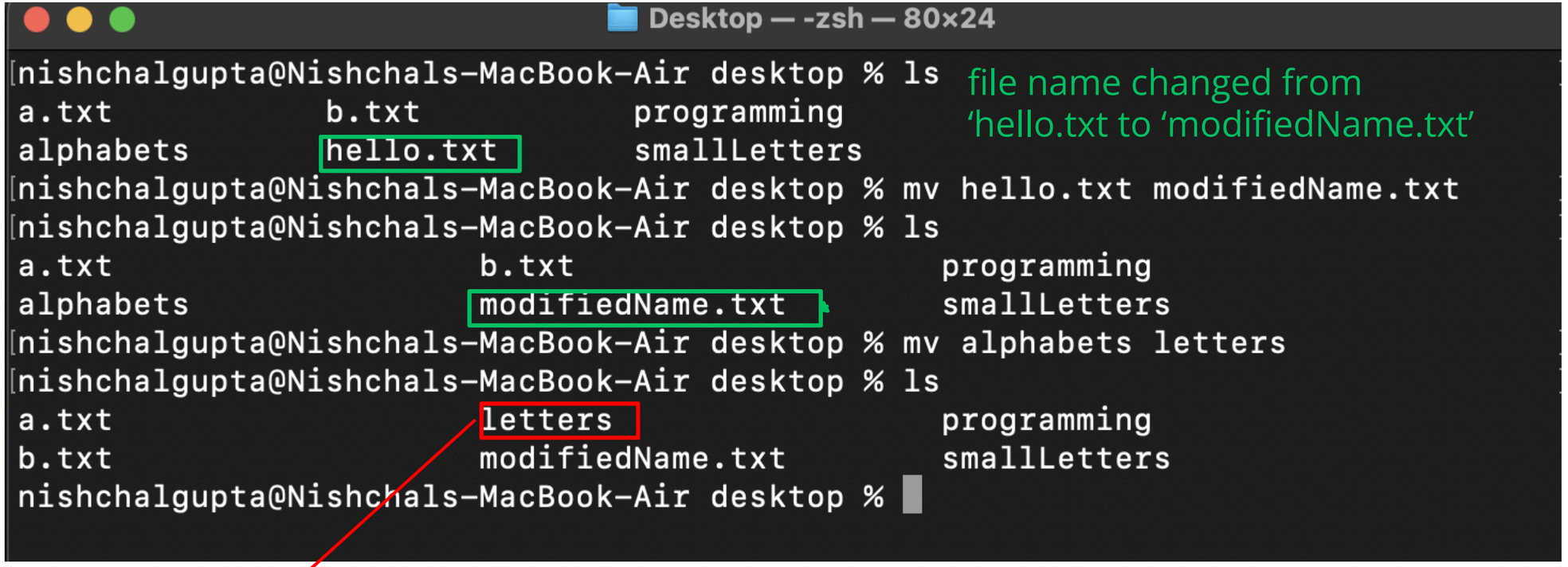


mv: Example: Rename Files and Folders



file name changed from 'hello.txt to 'modifiedName.txt'

folder name changed from 'alphabets' to 'letters'

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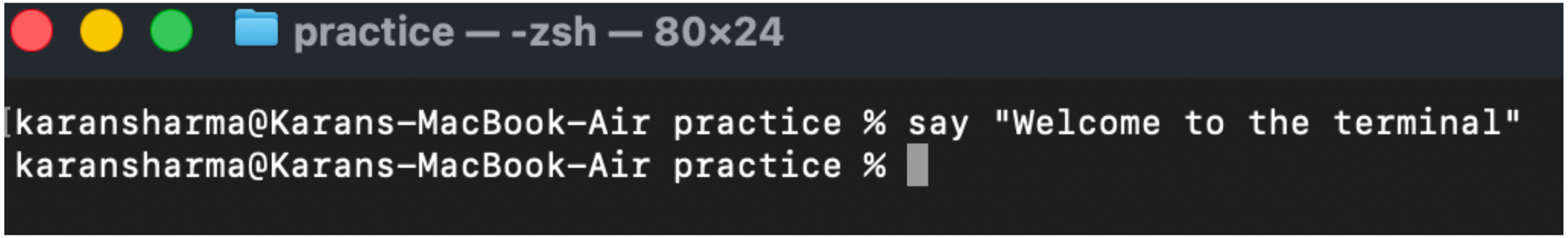
The fun part : Terminal Beyond Files

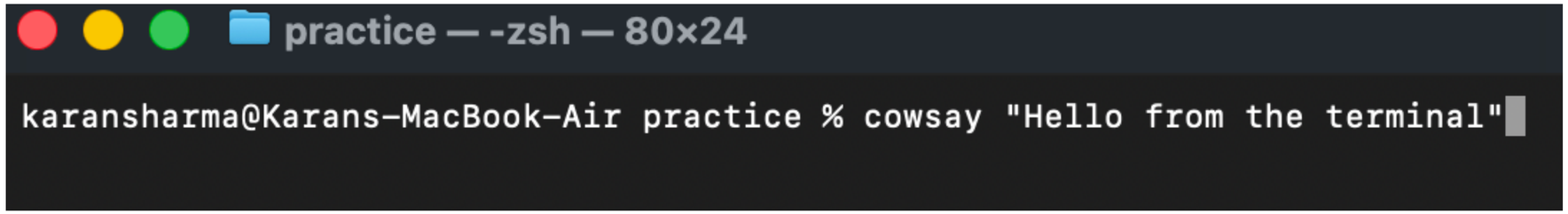
say : The machine talks

The say command makes your Mac speak the text aloud.

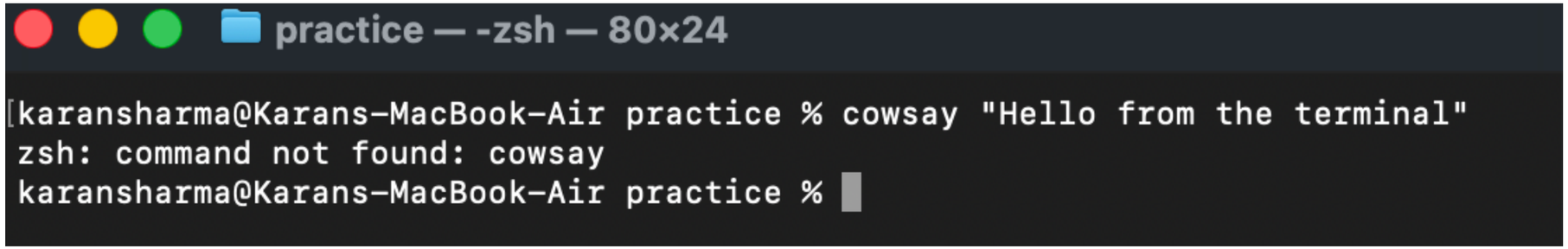


cowsay

A fun terminal command that displays messages using ASCII-art characters like cows, dragons, and penguins.



Why did this fail?

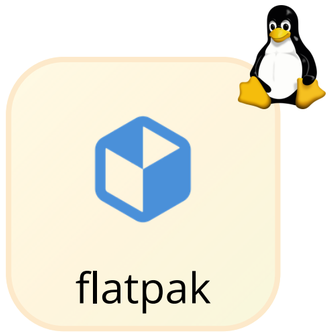


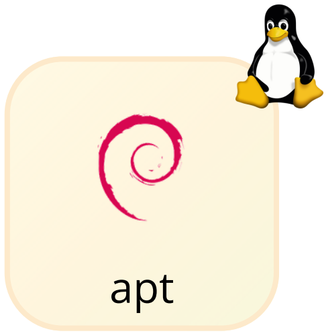
Because :

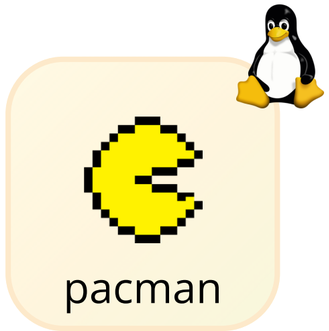
- A command is just a program
- say already comes with macOS
- cowsay is not installed yet
- The terminal can only run programs it can find

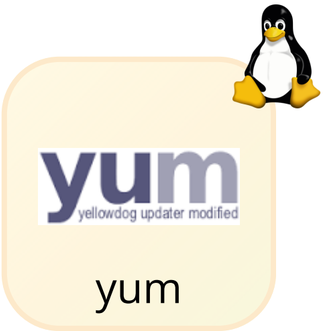
Package Management

Package management is the process of installing, updating, configuring, and removing software packages in an operating system using automated tools.









Package: *A package is a bundled collection of software components.*

So, Homebrew

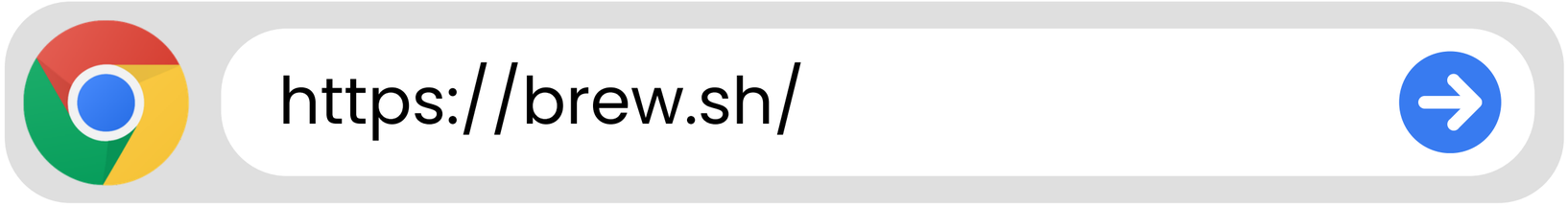
Homebrew is the package manager for macOS terminal.

It:

- finds programs
- downloads them
- installs them
- makes them available in terminal

Reference : Link

Installing homebrew




```
bash -c "$(curl -fsSL
https://raw.githubusercontent.com/Homebrew/
install/HEAD/install.sh)"
```

Installing fortune, cowsay, sl & lolcat

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```
brew install cowsay fortune sl lolcat
```

cowsay

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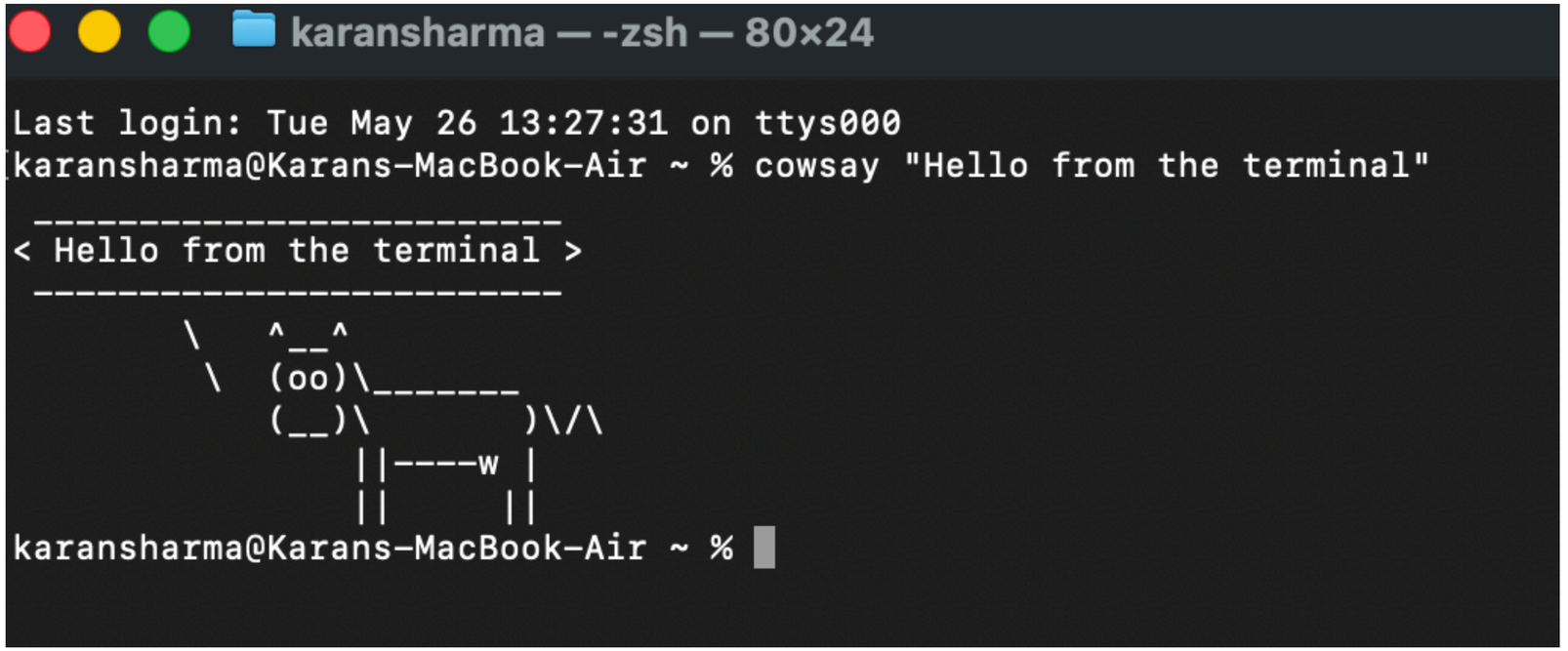


Commands Have Structure

cowsay -f dragon "I am a dragon"

cowsay → command

-f → option

dragon "I am a dragon" → argument

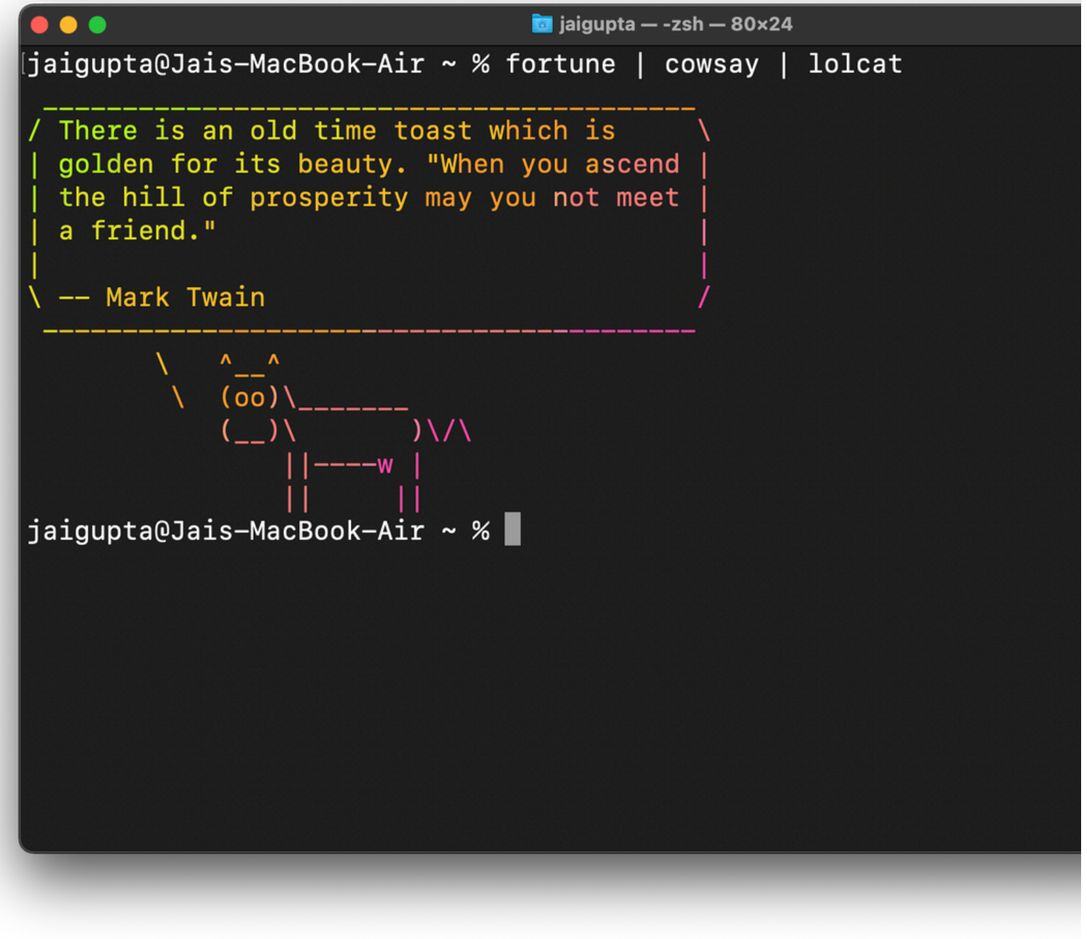
Options modify the behaviour of a command.

Piping Commands

```
fortune | cowsay
```

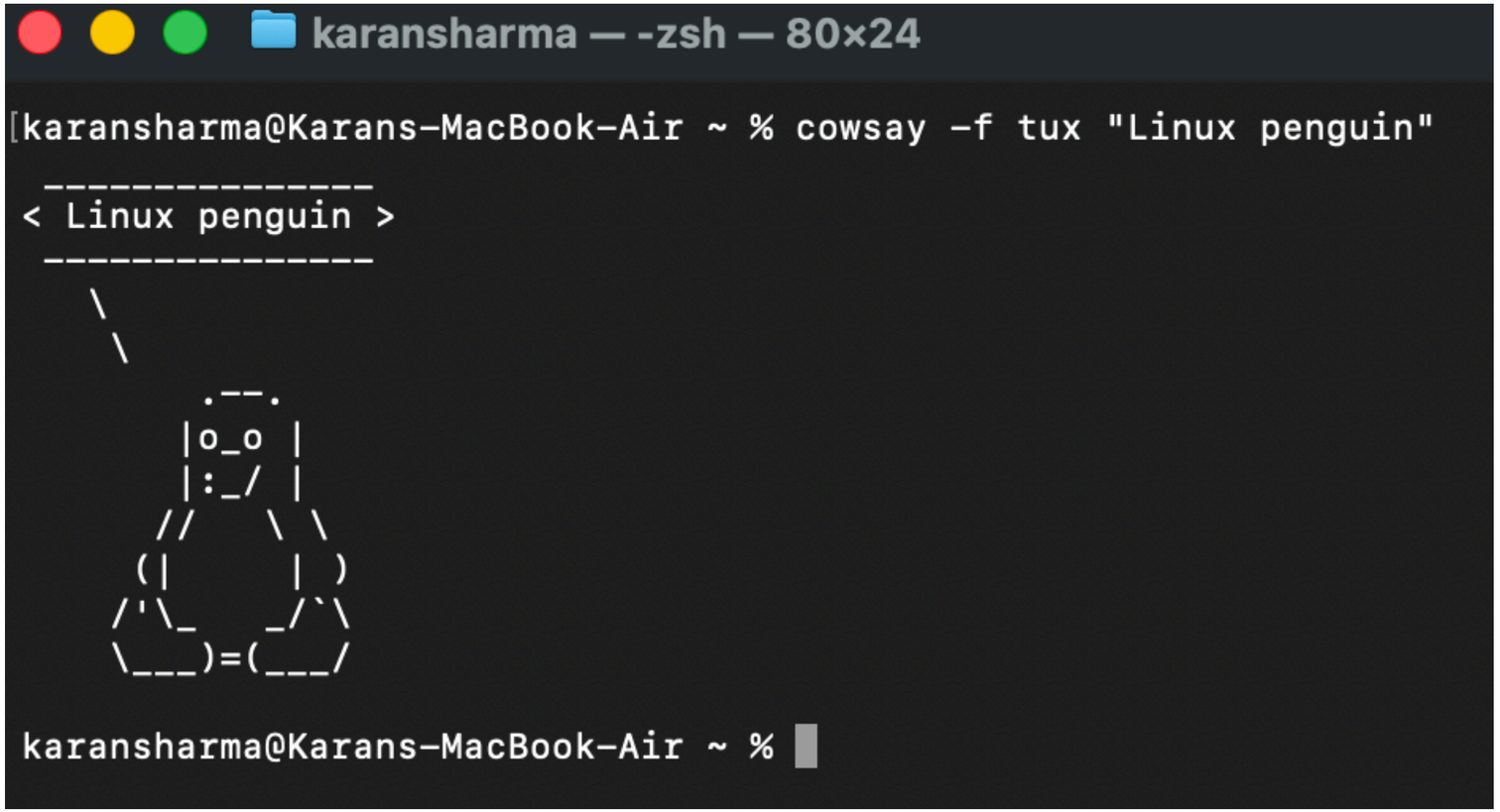
piping allows you to **pass output of one command as input** to another

```
fortune | cowsay | lolcat
```



Example

cowsay -f tux "Linux penguin"



Star Wars in the Terminal

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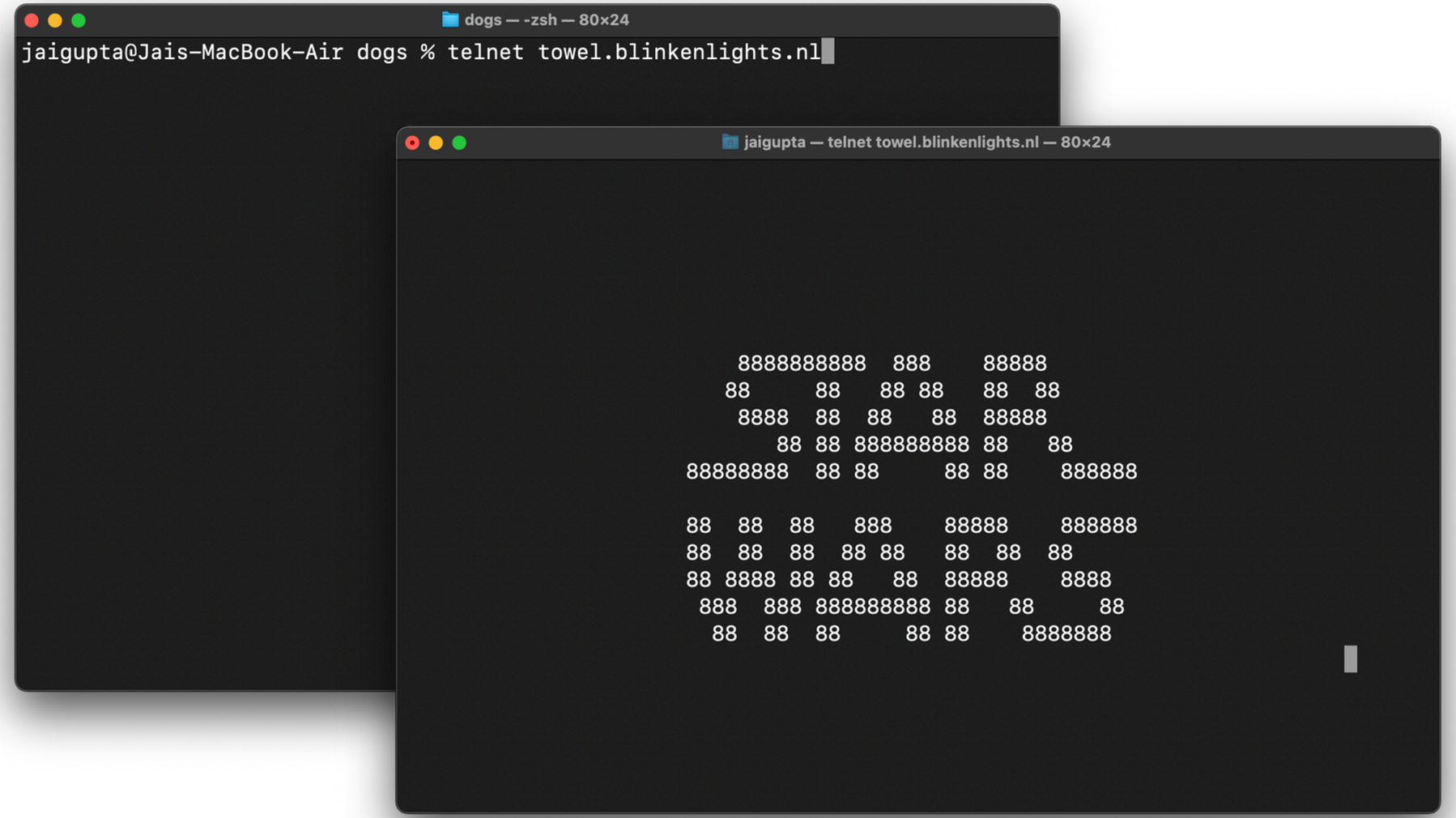
In 1997, Simon Jansen recreated Star Wars in ASCII art and made it watchable over the internet using telnet.

```
brew install telnet
```

```
telnet towel.blinkenlights.nl
```

Watch Star Wars: Screenshots



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Quiz Time

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Thank you!

*Share your valuable **feedback** on the portal!*